

43 Cheat Sheet

$$|a\rangle \triangleq \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} \quad a_0, a_1 \in \mathbb{C}$$

$$|a\rangle\langle a| = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} \begin{pmatrix} a_0^* & a_1^* \end{pmatrix} = \begin{pmatrix} a_0 a_0^* & a_0 a_1^* \\ a_1 a_0^* & a_1 a_1^* \end{pmatrix} = \begin{pmatrix} |a_0|^2 & a_0 a_1^* \\ a_1 a_0^* & |a_1|^2 \end{pmatrix}$$

$$\langle a|a\rangle = \begin{pmatrix} a_0^* & a_1^* \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = a_0^* a_0 + a_1^* a_1 = |a_0|^2 + |a_1|^2$$

2x2 matrix times 2x2 matrix

$$\begin{pmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{pmatrix} \begin{pmatrix} B_{1,1} & B_{1,2} \\ B_{2,1} & B_{2,2} \end{pmatrix} =$$

$$\begin{pmatrix} A_{1,1}B_{1,1} + A_{1,2}B_{2,1} & A_{1,1}B_{1,2} + A_{1,2}B_{2,2} \\ A_{2,1}B_{1,1} + A_{2,2}B_{2,1} & A_{2,1}B_{1,2} + A_{2,2}B_{2,2} \end{pmatrix}$$

3x1 matrix times 1x3 matrix

$$\begin{pmatrix} A_{1,1} \\ A_{2,1} \\ A_{3,1} \end{pmatrix} \begin{pmatrix} B_{1,1} & B_{1,2} & B_{1,3} \end{pmatrix} =$$

$$\begin{pmatrix} A_{1,1}B_{1,1} & A_{1,1}B_{1,2} & A_{1,1}B_{1,3} \\ A_{2,1}B_{1,1} & A_{2,1}B_{1,2} & A_{2,1}B_{1,3} \\ A_{3,1}B_{1,1} & A_{3,1}B_{1,2} & A_{3,1}B_{1,3} \end{pmatrix}$$

1x3 matrix times 3x1 matrix

$$\begin{pmatrix} A_{1,1} & A_{1,2} & A_{1,3} \end{pmatrix} \begin{pmatrix} B_{1,1} \\ B_{2,1} \\ B_{3,1} \end{pmatrix} =$$

$$A_{1,1}B_{1,1} + A_{1,2}B_{2,1} + A_{1,3}B_{3,1}$$

Transpose, adjoint

$$(A^T)_{i,j} = A_{j,i} \quad (A^\dagger)_{i,j} = A_{j,i}^*$$

Hermitian, Unitary Operators

$$A \text{ is Hermitian} \iff A^\dagger = A$$

$$U \text{ is Unitary} \iff U^\dagger U = U U^\dagger = I$$

Matrix Exponential

For a square matrix X ,

$$e^X \triangleq \sum_{n=0}^{\infty} \frac{X^n}{n!}$$

$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \dots$$

$$\frac{d}{dt} e^{At} = A e^{At} = e^{At} A$$

$U(t) = e^{tX}$ is the unique solution to $\frac{dU}{dt} = XU(t)$,
 $U(0) = I$.

If $X = PDP^{-1}$ with $D = \text{diag}(\lambda_1, \dots, \lambda_n)$,

$$e^X = P \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_n}) P^{-1}$$

If H is Hermitian, then e^{iH} is unitary:

$$(e^{iH})^\dagger e^{iH} = I$$

Spin-1/2, Pauli Matrices

This spin is for 2-dimensional Hilbert spaces. Electrons, protons, neutrons and quarks have spin-1/2. For spin-1/2 systems,

$$S_x \triangleq \frac{\hbar}{2} \sigma_x \quad S_y \triangleq \frac{\hbar}{2} \sigma_y \quad S_z \triangleq \frac{\hbar}{2} \sigma_z$$

$$S_x \triangleq \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad S_y \triangleq \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad S_z \triangleq \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$|z+\rangle = |\uparrow\rangle = |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |z-\rangle = |\downarrow\rangle = |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|x+\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$|x-\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$|y+\rangle = \frac{1}{\sqrt{2}}(|z+\rangle + i|z-\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$|y-\rangle = \frac{1}{\sqrt{2}}(|z+\rangle - i|z-\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

Spin-1

This spin is for 3-dimensional Hilbert spaces. W/Z bosons, photons and gluons have spin-1. For spin-1 systems,

$$S_x \triangleq \frac{\hbar}{2} \Sigma_x \quad S_y \triangleq \frac{\hbar}{2} \Sigma_y \quad S_z \triangleq \frac{\hbar}{2} \Sigma_z$$

$$S_x \triangleq \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad S_y \triangleq \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

$$S_z \triangleq \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad |0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad |-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

For photons, S_z is often called rectilinear and S_x is often called diagonal and S_y are often called circular.

Schrödinger Equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle$$

Uniform Dynamics

For time independent Hamiltonian $H(t) = H$,

$$\Rightarrow U(t, t_0) = e^{-iH(t-t_0)/\hbar}$$

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle$$

Born's Rule

The probability of getting the eigenvalue λ_α of eigenstate $|\alpha\rangle$ is

$$P_\alpha = |\langle \alpha | \psi \rangle|^2$$

Heisenberg Picture

$$\Delta A \Delta B \geq \frac{1}{2} \left| \mathbb{E} \left[[\hat{A}, \hat{B}] \right] \right|$$

$$\Delta A^2 \Delta B^2 \geq \left| \langle \psi | \frac{1}{2j} [\hat{A}, \hat{B}] | \psi \rangle \right|^2 = \left(\frac{1}{2j} \mathbb{E} \left[[\hat{A}, \hat{B}] \right] \right)^2$$

Ehrenfest Theorem

$$\frac{d}{dt} \mathbb{E}[A(t)] = \frac{i}{\hbar} \mathbb{E}[[H, A]]$$

EV, Variance, Standard Deviation

$$\mathbb{E}[X] = \sum_x x P(x)$$

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\text{Std}(X) = \sqrt{\text{Var}(X)}$$

Time Energy Uncertainty

$$\Delta E \cdot T \geq \hbar = \frac{h}{2}$$

Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_a (x-a)^n$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Euler's Formula

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$